

Human Activity Recognition on Video

Project Overview:

Human Activity Recognition (HAR) is a vital field in computer vision that involves identifying and classifying various human actions and behaviors in video data. This project aims to develop a deep learning-based model to recognize and classify human activities from video sequences. The chosen dataset, UCF50, is a standard action recognition dataset containing videos of 50 different human actions performed by various subjects in a wide variety of real-world scenarios.

Problem Selection:

The problem selected is human activity recognition from video data. This is an important and challenging task as it has applications in areas like surveillance, sports analytics and others. Accurate recognition of human actions can lead to improved security systems and enhanced sports analysis. This problem is particularly interesting because it requires the model to understand the spatiotemporal features of the actions, meaning it must capture not only the spatial arrangement of the objects but also their temporal evolution across video frames.

Dataset: UCF50 - Action Recognition Data Set

Proposed Architecture: A convolutional neural network (CNN) combined with a Long-Short Term Memory Network (LSTM).

Framework: Pytorch

Performance Evaluation:

The performance of the model will be judged using the following metrics:

Accuracy: The primary metric will be overall classification accuracy across the 50 action categories.

Confusion Matrix: To understand how well the model distinguishes between different classes and identify common misclassifications.